Acid–Base Disorders – Section Summary

HAGMA • NAGMA • Alkalosis • Mixed disorders • Toxins

Critical Care Board Review • Section 2 • MCQ-Oriented

1. Anion Gap Acidosis

- Toxic alcohol (visual symptoms) → **fomepizole + haemodialysis**
- Short bowel → **D-lactic acidosis**
- Salicylate → urine alkalinisation ($NaHCO_3$); mixed resp. alkalosis + HAGMA (+ metabolic alkalosis if vomiting)
- Metformin: bicarbonate may aid excretion; dialysis for severe

2. Non-gap & RTA

- Toluene (glue sniffing) → NAGMA (distal RTA), hypokalaemia
- Type 1 RTA: urine pH > 5.5, low K

3. Metabolic Alkalosis

- Vomiting → **H^+ and Cl^- loss**
- Hidden cause with ↑ urine Cl → **diuretics**
- Massive transfusion → **citrate → bicarbonate**

4. Respiratory & Mixed

- COPD: chronic respiratory acidosis with compensation, normal A–a
- Diuretic + COPD → mixed metabolic alkalosis + respiratory acidosis

5. High-Yield MCQ Pearls

- ☑ Massive transfusion → alkalosis
- ☑ Toxic alcohol → fomepizole + HD

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5. High-Yield MCQ Pearls

- [x] Massive transfusion → alkalosis
- [x] Toxic alcohol → fomepizole + HD